

**Vikavirtasuojien mittausraportti ja hätäseispainikkeiden testaus**

Tarkastuspäivä 6.5.2023

Tarkastaja Roope Ruhanen

Kohde Valmet Automation

Osallistujat Mika Helassalo

Osoite Lentokentäkatu 11 Tampere

| Ryhmä          | Vaihe          | Laukaisuvirta (mA) | Laukaisuaika (ms) | Painiketestaus | Lisätietoja |
|----------------|----------------|--------------------|-------------------|----------------|-------------|
| MOD 66: JKA-17 |                |                    |                   |                |             |
| 100            | L <sub>1</sub> | 21                 | 31                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 39                | X              |             |
| 101            | L <sub>1</sub> | 21                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 34                | X              |             |
| 102            | L <sub>1</sub> | 21                 | 31                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 30                | X              |             |
| 103            | L <sub>1</sub> | 21                 | 30                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 104            | L <sub>1</sub> | 27                 | 34                | X              |             |
|                | L <sub>2</sub> | 15                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 49                | X              |             |
| 105            | L <sub>1</sub> | 21                 | 30                | X              |             |
|                | L <sub>2</sub> | 21                 | 30                | X              |             |
|                | L <sub>3</sub> | 21                 | 30                | X              |             |
| 106            | L <sub>1</sub> | 21                 | 31                | X              |             |
|                | L <sub>2</sub> | 21                 | 32                | X              |             |
|                | L <sub>3</sub> | 24                 | 29                | X              |             |
| 107            | L <sub>1</sub> | 21                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 108            | L <sub>1</sub> | 24                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 30                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 109            | L <sub>1</sub> | 21                 | 30                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 30                | X              |             |
| 110            | L <sub>1</sub> | 21                 | 33                | X              |             |
|                | L <sub>2</sub> | 21                 | 30                | X              |             |
|                | L <sub>3</sub> | 24                 | 35                | X              |             |
| 111            | L <sub>1</sub> | 24                 | 36                | X              |             |
|                | L <sub>2</sub> | 21                 | 34                | X              |             |
|                | L <sub>3</sub> | 18                 | 53                | X              |             |
| 112            | L <sub>1</sub> | 21                 | 30                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 113            | L <sub>1</sub> | 21                 | 34                | X              |             |
|                | L <sub>2</sub> | 24                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 30                | X              |             |
| 114            | L <sub>1</sub> | 21                 | 30                | X              |             |
|                | L <sub>2</sub> | 21                 | 30                | X              |             |
|                | L <sub>3</sub> | 21                 | 30                | X              |             |
| 115            | L <sub>1</sub> | 21                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 30                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 116            | L <sub>1</sub> | 24                 | 33                | X              |             |
|                | L <sub>2</sub> | 21                 | 32                | X              |             |

|     |                |    |    |   |  |
|-----|----------------|----|----|---|--|
|     | L <sub>3</sub> | 21 | 36 | X |  |
| 117 | L <sub>1</sub> | 21 | 30 | X |  |
|     | L <sub>2</sub> | 21 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 29 | X |  |
| 118 | L <sub>1</sub> | 21 | 31 | X |  |
|     | L <sub>2</sub> | 21 | 31 | X |  |
|     | L <sub>3</sub> | 21 | 31 | X |  |
| 119 | L <sub>1</sub> | 21 | 29 | X |  |
|     | L <sub>2</sub> | 21 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 29 | X |  |
| 120 | L <sub>1</sub> | 21 | 38 | X |  |
|     | L <sub>2</sub> | 27 | 35 | X |  |
|     | L <sub>3</sub> | 18 | 31 | X |  |
| 121 | L <sub>1</sub> | 18 | 31 | X |  |
|     | L <sub>2</sub> | 21 | 34 | X |  |
|     | L <sub>3</sub> | 24 | 33 | X |  |
| 122 | L <sub>1</sub> | 21 | 30 | X |  |
|     | L <sub>2</sub> | 21 | 30 | X |  |
|     | L <sub>3</sub> | 21 | 30 | X |  |
| 123 | L <sub>1</sub> | 24 | 29 | X |  |
|     | L <sub>2</sub> | 24 | 29 | X |  |
|     | L <sub>3</sub> | 24 | 29 | X |  |
| 124 | L <sub>1</sub> | 21 | 29 | X |  |
|     | L <sub>2</sub> | 21 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 29 | X |  |
| 125 | L <sub>1</sub> | 21 | 31 | X |  |
|     | L <sub>2</sub> | 24 | 31 | X |  |
|     | L <sub>3</sub> | 21 | 31 | X |  |
| 126 | L <sub>1</sub> | 24 | 31 | X |  |
|     | L <sub>2</sub> | 24 | 30 | X |  |
|     | L <sub>3</sub> | 21 | 33 | X |  |
| 127 | L <sub>1</sub> | 21 | 36 | X |  |
|     | L <sub>2</sub> | 27 | 32 | X |  |
|     | L <sub>3</sub> | 21 | 30 | X |  |
| 128 | L <sub>1</sub> | 24 | 31 | X |  |
|     | L <sub>2</sub> | 21 | 31 | X |  |
|     | L <sub>3</sub> | 24 | 30 | X |  |
| 129 | L <sub>1</sub> | 21 | 32 | X |  |
|     | L <sub>2</sub> | 24 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 31 | X |  |
| 130 | L <sub>1</sub> | 21 | 30 | X |  |
|     | L <sub>2</sub> | 21 | 30 | X |  |
|     | L <sub>3</sub> | 21 | 31 | X |  |
| 131 | L <sub>1</sub> | 21 | 42 | X |  |
|     | L <sub>2</sub> | 24 | 36 | X |  |
|     | L <sub>3</sub> | 24 | 31 | X |  |
| 132 | L <sub>1</sub> | 24 | 29 | X |  |
|     | L <sub>2</sub> | 24 | 29 | X |  |
|     | L <sub>3</sub> | 24 | 29 | X |  |
| 133 | L <sub>1</sub> | 21 | 29 | X |  |
|     | L <sub>2</sub> | 21 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 29 | X |  |
| 134 | L <sub>1</sub> | 21 | 38 | X |  |
|     | L <sub>2</sub> | 21 | 49 | X |  |
|     | L <sub>3</sub> | 24 | 31 | X |  |
| 135 | L <sub>1</sub> | 24 | 31 | X |  |
|     | L <sub>2</sub> | 21 | 31 | X |  |
|     | L <sub>3</sub> | 21 | 30 | X |  |
| 136 | L <sub>1</sub> | 21 | 31 | X |  |
|     | L <sub>2</sub> | 24 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 32 | X |  |

|     |                |    |    |   |  |
|-----|----------------|----|----|---|--|
| 137 | L <sub>1</sub> | 21 | 30 | X |  |
|     | L <sub>2</sub> | 21 | 30 | X |  |
|     | L <sub>3</sub> | 21 | 30 | X |  |
| 143 | L <sub>1</sub> | 21 | 30 | X |  |
|     | L <sub>2</sub> | 21 | 30 | X |  |
|     | L <sub>3</sub> | 21 | 30 | X |  |
| 144 | L <sub>1</sub> | 21 | 29 | X |  |
|     | L <sub>2</sub> | 21 | 29 | X |  |
|     | L <sub>3</sub> | 21 | 29 | X |  |
| 203 | L <sub>1</sub> | 24 | 18 | X |  |
| 204 | L <sub>1</sub> | 21 | 28 | X |  |
| 205 | L <sub>1</sub> | 21 | 28 | X |  |
| 206 | L <sub>1</sub> | 24 | 18 | X |  |
| 207 | L <sub>1</sub> | 24 | 38 | X |  |
| 208 | L <sub>1</sub> | 24 | 18 | X |  |
| 209 | L <sub>1</sub> | 24 | 38 | X |  |
| 210 | L <sub>1</sub> | 24 | 18 | X |  |
| 211 | L <sub>1</sub> | 21 | 29 | X |  |
| 212 | L <sub>1</sub> | 21 | 29 | X |  |
| 213 | L <sub>1</sub> | 27 | 39 | X |  |
| 214 | L <sub>1</sub> | 24 | 38 | X |  |
| 215 | L <sub>1</sub> | 21 | 29 | X |  |
| 223 | L <sub>1</sub> | 24 | 28 | X |  |
| 224 | L <sub>1</sub> | 24 | 19 | X |  |
| 225 | L <sub>1</sub> | 24 | 38 | X |  |
| 226 | L <sub>1</sub> | 24 | 19 | X |  |
| 227 | L <sub>1</sub> | 24 | 38 | X |  |
| 228 | L <sub>1</sub> | 21 | 28 | X |  |
| 229 | L <sub>1</sub> | 24 | 18 | X |  |
| 230 | L <sub>1</sub> | 21 | 39 | X |  |
| 231 | L <sub>1</sub> | 24 | 19 | X |  |
| 232 | L <sub>1</sub> | 24 | 27 | X |  |
| 233 | L <sub>1</sub> | 24 | 18 | X |  |
| 234 | L <sub>1</sub> | 24 | 38 | X |  |
| 235 | L <sub>1</sub> | 24 | 39 | X |  |

#### Hätäseispainikkeet

Keskuksen hätäseispainikkeiden toiminta testattu, painikkeiden toiminta moitteeton.

#### Lisätietoa

Mitatut vikavirtasuojat tai vikavirtajohdonsuojakatkaisijat ovat A-tyyppiä 30 mA nimellisvirralla

Mittaukset suoritettu nousevalla vikavirralla, tämä vaikuttaa laukaisuaikoihin

Käytetty mittalaite: Amprobe Telaris 0100-Plus

Varalla olevia vikavirtasuojia tai vikavirtajohdonsuojakatkaisijoita ei mitattu



## Vikavirtasuojien mittausraportti ja hätäseispainikkeiden testaus

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| Ryhmä          | Vaihe          | Laukaisuvirta (mA) | Laukaisuaika (ms) | Painiketestaus | Lisätietoja |
|----------------|----------------|--------------------|-------------------|----------------|-------------|
| MOD 66: JKA-18 |                |                    |                   |                |             |
| 100            | L <sub>1</sub> | 24                 | 43                | X              |             |
|                | L <sub>2</sub> | 15                 | 32                | X              |             |
|                | L <sub>3</sub> | 18                 | 35                | X              |             |
| 101            | L <sub>1</sub> | 21                 | 49                | X              |             |
|                | L <sub>2</sub> | 24                 | 44                | X              |             |
|                | L <sub>3</sub> | 24                 | 33                | X              |             |
| 102            | L <sub>1</sub> | 24                 | 32                | X              |             |
|                | L <sub>2</sub> | 21                 | 30                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 103            | L <sub>1</sub> | 21                 | 36                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 104            | L <sub>1</sub> | 21                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 105            | L <sub>1</sub> | 21                 | 31                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 106            | L <sub>1</sub> | 21                 | 30                | X              |             |
|                | L <sub>2</sub> | 21                 | 37                | X              |             |
|                | L <sub>3</sub> | 24                 | 29                | X              |             |
| 107            | L <sub>1</sub> | 21                 | 32                | X              |             |
|                | L <sub>2</sub> | 21                 | 33                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 108            | L <sub>1</sub> | 24                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 32                | X              |             |
|                | L <sub>3</sub> | 21                 | 32                | X              |             |
| 109            | L <sub>1</sub> | 21                 | 34                | X              |             |
|                | L <sub>2</sub> | 24                 | 30                | X              |             |
|                | L <sub>3</sub> | 21                 | 32                | X              |             |
| 110            | L <sub>1</sub> | 21                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 118            | L <sub>1</sub> | 21                 | 29                | X              |             |
|                | L <sub>2</sub> | 21                 | 29                | X              |             |
|                | L <sub>3</sub> | 21                 | 29                | X              |             |
| 119            | L <sub>1</sub> | 21                 | 31                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 120            | L <sub>1</sub> | 21                 | 31                | X              |             |
|                | L <sub>2</sub> | 21                 | 31                | X              |             |
|                | L <sub>3</sub> | 21                 | 31                | X              |             |
| 206            | L <sub>1</sub> | 24                 | 18                | X              |             |
| 207            | L <sub>1</sub> | 24                 | 18                | X              |             |
| 208            | L <sub>1</sub> | 21                 | 28                | X              |             |
| 209            | L <sub>1</sub> | 24                 | 18                | X              |             |
| 210            | L <sub>1</sub> | 21                 | 29                | X              |             |
| 211            | L <sub>1</sub> | 24                 | 18                | X              |             |
| 212            | L <sub>1</sub> | 24                 | 18                | X              |             |
| 213            | L <sub>1</sub> | 21                 | 28                | X              |             |

|     |                |    |    |   |  |
|-----|----------------|----|----|---|--|
| 214 | L <sub>1</sub> | 24 | 18 | X |  |
| 215 | L <sub>1</sub> | 21 | 28 | X |  |
| 216 | L <sub>1</sub> | 21 | 19 | X |  |
| 217 | L <sub>1</sub> | 24 | 18 | X |  |
| 219 | L <sub>1</sub> | 24 | 27 | X |  |
| 220 | L <sub>1</sub> | 21 | 29 | X |  |

#### Hätäseispainikkeet

Keskuksen hätäseispainikkeiden toiminta testattu, painikkeiden toiminta moitteeton.

#### Lisätietoa

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Mittaukset suoritettu nousevalla vikavirralla, tämä vaikuttaa laukaisuaikoihin

Käytetty mittalaite: Amprobe Telaris 0100-Plus

Varalla olevia vikavirtasuojia tai vikavirtajohdonsuojakatkaisijoita ei mitattu